Chapter 4
SAMPLING DISTRIBUTIONS AND POIN ESTIMATION OF PARAMETERS

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CHAPTER OUTLINE

Chapter outline

LEARNING OBJECTIVES

Learning Objectives

After careful study of this chapter you should be able to do the following:

1. Explain the general concepts of estimating the parameters of a population or a probability distribution
2. Explain the important role of the normal distribution as a sampling distribution
3. Understand the central limit theorem
4. Explain important properties of point estimators, including bias, variance, and mean square error
5. Know how to construct point estimators using the method of moments and the method of maximum likelihood (read only)
6. Know how to compute and explain the precision with which a parameter is estimated (read only)
7. Know how to construct a point estimator using the Bayesian approach (read only)

Content

Populations

Definition 1 (Population)

A population consists of the totality of the observations with which we are concerned.

Size

The number of observations in the population is defined to be the size of the population N.

Populations

Example 1

1. If there are 600 students in the school whom we classified according to blood type, we say that we have a population of size $N = 600$.
2. The numbers on the cards in a deck, the heights of residents in a certain city, and the lengths of fish in a particular lake are examples of populations with finite size. In each case, the total number of observations is a finite number.
3. The observations obtained by measuring the atmospheric pressure every day, from the past on into the future, or all measurements of the depth of a lake, from any conceivable position, are examples of populations whose sizes are infinite.

Samples

Definition 2 (Sample)

A sample is a subset of a population.

Size

The number of observations in the sample is defined to be the size of the sample n.

Random Sample

Definition 3 (Random sample)

Let $X_1, X_2, \ldots, X_n$ be n independent random variables, each having the same probability distribution $F_X(x)$. Define $(X_1, X_2, \ldots, X_n)$ to be a random sample of size n from the population $F_X(x)$ and write its joint probability distribution as

$$\boxed{F_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n) = F_{X_1}(x_1)F_{X_2}(x_2)\ldots F_{X_n}(x_n)} \tag{1}$$

Note

The random variables $X_1, X_2, \ldots, X_n$ are a random sample of size n if

(a) the X_i's are independent random variables, and

(b) every X_i has the same probability distribution.

Random Sample

Example 2

If one makes a random selection of $n = 8$ storage batteries from a manufacturing process that has maintained the same specification throughout and records the length of life for each battery, with the first measurement x_1 being a value of X_1, the second measurement x_2 a value of X_2, and so forth, then $x_1, x_2, \dots, x_8$ are the values of the random sample $X_1, X_2, \dots, X_8$. If we assume the population of battery lives to be normal, the possible values of any X_i, $i = 1, 2, \dots, 8$, will be precisely the same as those in the original population, and hence X_i has the same identical normal distribution as X.

Some Important Statistics

Definition 4 (Statistic)

Any function of the random variables constituting a random sample is called a statistic.

Example 3

For example, if $(X_1, X_2, \ldots, X_n)$ is a random sample of size n,

- the sample mean $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$,
- the sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$, and
- the sample standard deviation $S = \sqrt{S^2}$

are statistics.

Sample Mean. Median. Mode

Definition 5 (Sample mean)

- Sample mean: Let $(X_1, X_2, \ldots, X_n)$ be a random sample of size n,

$$\boxed{\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i} \tag{2}$$

Note that the statistic $\overline{X}$ assumes the value $\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ when X_1 assumes the value x_1, X_2 assumes the value x_2, and so forth.

- Sample median:

$$\hat{x} = \begin{cases} x_{(n+1)/2}, & \text{if } n \text{ is odd,} \\ \frac{1}{2}(x_{n/2} + x_{n/2+1}), & \text{if } n \text{ is even.} \end{cases}$$

- The sample mode is the value of the sample that occurs most often.

Sample Mean

Example 4

Suppose a data set consists of the following observations:

$$0.32 \quad 0.53 \quad 0.28 \quad 0.37 \quad 0.47 \quad 0.43 \quad 0.36 \quad 0.42 \quad 0.38 \quad 0.43.$$

The sample mode is 0.43, since this value occurs more than any other value.

Sample Variance. Standard Deviation

Definition 6 (Sample variance, standard deviation, and range)

- Sample variance: Let $(X_1, X_2, \ldots, X_n)$ be a random sample of size n,

$$\boxed{S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2} \tag{3}$$

The computed value of S^2 for a given sample is denoted by $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$.

- Sample standard deviation:

$$\boxed{S = \sqrt{S^2}} \tag{4}$$

- Let $X_{\max}$ denote the largest of the X_i values and $X_{\min}$ the smallest. Sample range:

$$\boxed{\mathbf{R} = X_{\max} - X_{\min}} \tag{5}$$

Sample Variance. Standard Deviation

Example 5

A comparison of coffee prices at 4 randomly selected grocery stores in San Diego showed increases from the previous month of 12, 15, 17, and 20 cents for a 1-pound bag. Find the variance of this random sample of price increases.

Sample Variance. Standard Deviation. Range

Solution

Calculating the sample mean, we get

$$\overline{x} = \frac{12 + 15 + 17 + 20}{4} = 16 \quad \text{cents.}$$

Therefore,

$$\begin{aligned} s^2 &= \frac{1}{3} \sum_{i=1}^{4} (x_i - 16)^2 \\ &= \frac{(12-16)^2 + (15-16)^2 + (17-16)^2 + (20-16)^2}{3} \\ &= \frac{34}{3}. \end{aligned}$$

Sample Variance. Standard Deviation

Theorem 1

If S^2 is the variance of a random sample of size n, we may write

$$\boxed{S^2 = \frac{1}{n(n-1)}\left[n\sum_{i=1}^{n} X_i^2 - \left(\sum_{i=1}^{n} X_i\right)^2\right]} \tag{6}$$

Sample Variance. Standard Deviation

Example 6

Find the variance of the data 3, 4, 5, 6, 6, and 7, representing the number of trout caught by a random sample of 6 fishermen on June 19, 1996, at Lake Muskoka.

Sample Variance. Standard Deviation

Solution

We find that $\sum_{i=1}^{6} x_i^2 = 171$, $\sum_{i=1}^{6} x_i = 311$, and $n = 6$. Hence,

$$s^2 = \frac{1}{(6)(5)}[(6)(171) - (31)^2] = \frac{13}{6}.$$

Thus, the sample standard deviation $s = \sqrt{13/6} = 1.47$ and the sample range is $7 - 3 = 4$.

Sample Variance. Standard Deviation

Note

Sample	Population
n: number of measurements in the sample	N: number of measurements in the population
$\overline{x}$: sample mean	μ: population mean
s^2: sample variance	σ^2: population variance
s: sample standard deviation	σ: population standard deviation

Sample Variance. Standard Deviation

Calculating Mean and Standard Deviation Using 570 VN Plus

The following commands show how to calculate the mean and standard deviation by using the STAT mode on a CASIO FX 570 VN PLUS (similar for other CASIO models).

Steps

1. Enter the data
   - **MODE** $\rightarrow$ **3** $\rightarrow$ **AC**
   - **SHIFT** $\rightarrow$ **MODE** $\rightarrow \Downarrow \rightarrow$ **4(STAT)** $\rightarrow$ **1(ON)**
   - **SHIFT** $\rightarrow$ **1** $\rightarrow$ **1(TYPE)** $\rightarrow$ **1(1-VAR)**

     To finish entering the data press the AC button.
2. Calculating numerical summaries
   - The sample mean $\overline{x}$: **SHIFT** $\rightarrow$ **1** $\rightarrow$ **4(VAR)** $\rightarrow$ **2**
   - The sample standard deviation s: **SHIFT** $\rightarrow$ **1** $\rightarrow$ **4** $\rightarrow$ **4**

Sample Variance. Standard Deviation

Example 7

Find the mean and standard deviation of the following distribution

Variable (x_i)	20	21	22	23	24
Frequency (n_i)	5	8	11	10	6

Solution

22.1 and 1.2567.

Content

Introduction

Introduction

- Statistical inference is always focused on drawing conclusions about one or more parameters of a population. An important part of this process is obtaining estimates of the parameters.
- Suppose that we want to obtain a point estimate (a reasonable value) of a population parameter. We know that before the data are collected, the observations are considered to be random variables, say, $X_1, X_2, \ldots, X_n$. Therefore, any function of the observation, or any statistic, is also a random variable.
- For example, the sample mean $\overline{X}$ and the sample variance S^2 are statistics and they are also random variables.
- Since a statistic is a random variable, it has a probability distribution. We call the probability distribution of a statistic a sampling distribution.

Introduction

Introduction

- When discussing inference problems, it is convenient to have a general symbol to represent the parameter of interest. We will use the Greek symbol θ (theta) to represent the parameter.
- In general, if X is a random variable with probability distribution $P_X(x)$ or $f_X(x)$, characterized by the unknown parameter θ, and if $(X_1, X_2, \ldots, X_n)$ is a random sample of size n from X, the statistic

$$\widehat{\Theta} = h(X_1, X_2, \ldots, X_n)$$

  is called a point estimator of θ.
- Note that $\widehat{\Theta}$ is a random variable because it is a function of random variables.
- After the sample has been selected, $\widehat{\Theta}$ takes on a particular numerical value $\hat{\theta}$ called the point estimate of θ.

Point Estimation

Definition 7 (Point Estimator)

A point estimate of some population parameter θ is a single numerical value $\widehat{\theta}$ of a statistic $\widehat{\Theta}$. The statistic $\widehat{\Theta}$ is called the point estimator.

Example 8

Suppose that the random variable X is normally distributed with an unknown mean μ. The sample mean is a point estimator of the unknown population mean μ. That is, $\widehat{\mu} = \overline{X}$. After the sample has been selected, the numerical value $\overline{x}$ is the point estimate of μ. Thus, if $x_1 = 25$, $x_2 = 30$, $x_3 = 29$, and $x_4 = 31$ the point estimate of μ is

$$\overline{x} = \frac{25 + 30 + 29 + 31}{4} = 28.75.$$

Point Estimation

Example 9

Similarly, if the population variance σ^2 is also unknown, a point estimator for σ^2 is the sample variance S^2, and the numerical value $s^2 = 6.9$ calculated from the sample data is called the point estimate of σ^2 (see Example 8).

Note

Estimation problems occur frequently in engineering. We often need to estimate

- The mean μ of a single population.
- The variance σ^2 (or standard deviation σ) of a single population.
- The proportion p of items in a population that belong to a class of interest
- The difference in means of two populations, $\mu_1 - \mu_2$.
- The difference in two population proportions, $p_1 - p_2$.

Point Estimation

Note (continuous)

Reasonable point estimates of these parameters are as follows:

- For μ, the estimate is $\widehat{\mu} = \overline{x}$, the sample mean.
- For σ^2, the estimate is $\widehat{\sigma}^2 = s^2$, the sample variance.
- For p, the estimate is $\widehat{p} = x/n$, the sample proportion, where x is the number of items in a random sample of size n that belong to the class of interest.
- For $\mu_1 - \mu_2$, the estimate is $\widehat{\mu}_1 - \widehat{\mu}_2 = \overline{x}_1 - \overline{x}_2$, the difference between the sample means of two independent random samples.
- For $p_1 - p_2$, the estimate is $\widehat{p}_1 - \widehat{p}_2$, the difference between two sample proportions computed from two independent random samples.

Content

Sampling Distributions

Since a statistic is a random variable, it has a probability distribution.

Definition 8

The probability distribution of a statistic is called a sampling distribution.

Note

For example, the probability distribution of $\overline{X}$ is called the sampling distribution of the mean. The sampling distribution of a statistic depends on the distribution of the population, the size of the sample, and the method of sample selection.

Sampling Distribution of the Mean

Sampling Distribution of the Mean

The first important sampling distribution to be considered is that of the mean $\overline{X}$.

- Suppose that a random sample of n observations is taken from a normal population with mean μ and variance σ^2. Now each observation in this sample, say, $(X_1, X_2, \ldots, X_n)$, is a normally and independently distributed random variable with mean μ and variance σ^2.
- Then, because linear functions of independent, normally distributed random variables are also normally distributed, we conclude that the sample mean

$$\overline{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$$

has a normal distribution with mean

$$\mu_{\overline{X}} = \frac{\mu + \cdots + \mu}{n} = \mu$$

and variance

$$\sigma^2_{\overline{X}} = \frac{\sigma^2 + \cdots + \sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Sampling Distribution of the Mean

Example 10

An electronics company manufactures resistors that have a mean resistance of 100 ohms and a standard deviation of 10 ohms. The distribution of resistance is normal. Find the probability that a random sample of $n = 25$ resistors will have an average resistance less than 95 ohms.

Solution

Note that the sampling distribution of $\overline{X}$ is normal, with mean $\mu_{\overline{X}} = 100$ ohms and a standard deviation of $\sigma_{\overline{X}} = \sigma/\sqrt{n} = 10/\sqrt{25} = 2$. Therefore,

$$P(\overline{X} < 95) = \Phi\left(\frac{95 - 100}{2}\right) = \Phi(-2.5) = 1 - \Phi(2.5) \approx 1 - 0.99379 = 0.00621,$$

where, $\Phi(2.5) = 0.99379$.

Sampling Distribution of the Mean

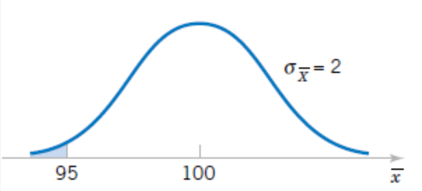

Figure 1: Probability for Example 10

Central Limit Theorem

Note

If we are sampling from a population that has an unknown probability distribution, the sampling distribution of the sample mean will still be approximately normal with mean μ and variance σ^2, if the sample size n is large. This is one of the most useful theorems in statistics, called the central limit theorem.

Theorem 2 (Central Limit Theorem)

If $(X_1, X_2, \dots, X_n)$ is a random sample of size n taken from a population (either finite or infinite) with mean μ and finite variance σ^2, and if $\overline{X}$ is the sample mean, the limiting form of the distribution of

$$\boxed{Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}} \tag{7}$$

as $n \to \infty$, is the standard normal distribution $\mathcal{N}(0, 1)$.

Central Limit Theorem

Example 11

The normal approximation for depends on the sample size n. The following Figures show the distributions of average scores obtained when tossing one, two, three, and five dice, respectively.

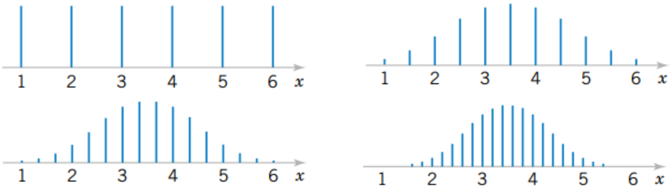

Figure 2: Distributions of average scores from throwing dice

Central Limit Theorem

Example 12

Suppose that a random variable X has a continuous uniform distribution

$$f_X(x) = \begin{cases} \frac{1}{2} & \text{nếu } x \in [4,6], \\ 0 & \text{otherwise.} \end{cases}$$

Find the distribution of the sample mean of a random sample of size $n = 40$.

Solution

The mean and variance of X are

$$\mu = \frac{4+6}{2} = 5; \quad \sigma^2 = \frac{(6-4)^2}{12} = \frac{1}{3}.$$

The central limit theorem indicates that the distribution of $\overline{X}$ is approximately normal with mean

$$\mu_{\overline{X}} = 5; \quad \sigma^2_{\overline{X}} = \frac{\sigma^2}{n} = \frac{1}{3(40)} = \frac{1}{120}.$$

Central Limit Theorem

The distributions of X and $\overline{X}$ are shown in the following Fig.

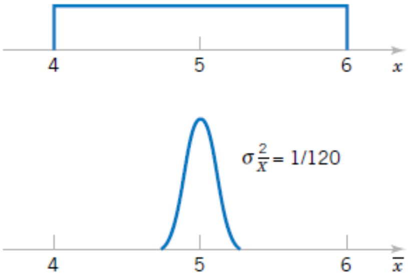

Figure 3: The distributions of X and $\overline{X}$ for Example 12

Central Limit Theorem

Note

1. The normal approximation for $\overline{X}$ will generally be good if $n \geq 30$, provided the population distribution is not terribly skewed.
2. If $n < 30$, the approximation is good only if the population is not too different from a normal distribution and, as stated above, if the population is known to be normal, the sampling distribution of $\overline{X}$ will follow a normal distribution exactly, no matter how small the size of the samples.

Central Limit Theorem

Note

The following Figure illustrates how the theorem works. It shows how the distribution of $\overline{X}$ becomes closer to normal as n grows larger, beginning with the clearly nonsymmetric distribution of an individual observation $(n = 1)$.

Central Limit Theorem

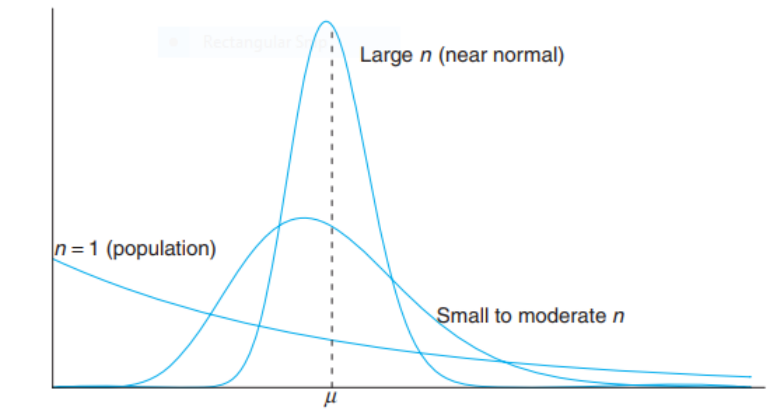

Figure 4: Illustration of the Central Limit Theorem (distribution of $\overline{X}$ for $n = 1$,moderate n, and large n)

Central Limit Theorem

Example 13

An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed, with a mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a random sample of 16 bulbs will have an average life of less than 775 hours.

Central Limit Theorem

Solution

The sampling distribution of $\overline{X}$ will be approximately normal, with $\mu_{\overline{X}} = 800$ and $\sigma_{\overline{X}} = 40/\sqrt{16} = 10$. The desired probability is given by the area of the shaded region in the following figure. Corresponding to $\overline{x} = 775$, we find that $z = \frac{775-800}{10} = -2.5$ and therefore

$$P(\overline{X} < 775) = P(Z < -2.5) = 0.0062.$$

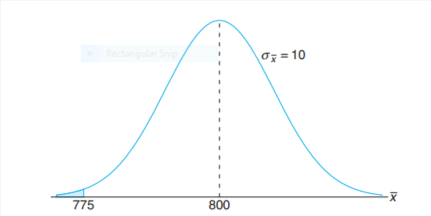

Figure 5: Area for Example 13

Table

Table 1 $\left(\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{\frac{-t^2}{2}} dt\right)$

x	0	1	2	3	4	5	6	7	8	9
0.0	0.50000	50399	50798	51197	51595	51994	52392	52790	53188	53586
0.1	53983	54380	54776	55172	55567	55962	56356	56749	57142	57535
0.2	57926	58317	58706	59095	59483	59871	60257	60642	61026	61409
0.3	61791	62172	62556	62930	63307	63683	64058	64431	64803	65173
0.4	65542	65910	66276	66640	67003	67364	67724	68082	68439	68739
0.5	69146	69447	69847	70194	70544	70884	71226	71566	71904	72240
0.6	72575	72907	73237	73565	73891	74215	74537	74857	75175	75490
0.7	75804	76115	76424	76730	77035	77337	77637	77935	78230	78524
0.8	78814	79103	79389	79673	79955	80234	80511	80785	81057	81327
0.9	81594	81859	82121	82381	82639	82894	83147	83398	83646	83891
1.0	84134	84375	84614	84850	85083	85314	85543	85769	85993	86214
1.1	86433	86650	86864	87076	87286	87493	87698	87900	88100	88298
1.2	88493	88686	88877	89065	89251	89435	89617	89796	89973	90147
1.3	90320	90490	90658	90824	90988	91149	91309	91466	91621	91774
1.4	91924	92073	92220	92364	92507	92647	92786	92922	93056	93189
1.5	93319	93448	93574	93699	93822	93943	94062	94179	94295	94408
1.6	94520	94630	94738	94845	94950	95053	95154	95254	95352	95449
1.7	95543	95637	95728	95818	95907	95994	96080	96164	96246	96327
1.8	96407	96485	96562	96638	96712	96784	96856	96926	96995	97062
1.9	97128	97193	97257	97320	97381	97441	**97500**	97558	97615	97670

Table 1 (continuous)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{\frac{-t^2}{2}} dt$$

x	0	1	2	3	4	5	6	7	8	9
2.0	97725	97778	97831	97882	97932	97982	98030	98077	98124	98169
2.1	98214	98257	98300	98341	98382	98422	99461	98500	98537	98574
2.2	98610	98645	98679	98713	98745	98778	98809	98840	98870	98899
2.3	98928	98956	98983	99010	99036	99061	99086	99111	99134	99158
2.4	99180	99202	99224	99245	99266	99285	99305	99324	99343	99361
2.5	99379	99396	99413	99430	99446	99261	99477	99492	99506	99520
2.6	99534	99547	99560	99573	99585	99598	99609	99621	99632	99643
2.7	99653	99664	99674	99683	99693	99702	99711	99720	99728	99763
2.8	99744	99752	99760	99767	99774	99781	99788	99795	99801	99807
2.9	99813	99819	99825	99831	99836	99841	99846	99851	99856	99861
3.0	0,99865	3,1	99903	3,2	99931	3,3	99952	3,4	99966	
3.5	99977	3,6	99984	3,7	99989	3,8	99993	3,9	99995	
4.0	999968									
4.5	999997									
5.0	99999997									

χ^2 Distribution

Theorem 3 (χ^2 Distribution)

If S^2 is the variance of a random sample of size n taken from a normal population having the variance σ^2, then the statistic

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n} \frac{(X_i - \overline{X})^2}{\sigma^2}$$

has a chi-squared distribution with $\nu = n - 1$ degrees freedom.

PDF

The probability density function of a χ^2 random variable is

$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{(k/2)-1} e^{-e/2} \quad x > 0 \tag{8}$$

where k is the number of degrees of freedom. The mean and variance of the χ^2 distribution are k and $2k$, respectively. Several chi-square distributions are shown in Fig. 6. Note that the chi-square random variable is nonnegative and that the probability distribution is skewed to the right. However, as k increases, the distribution becomes more symmetric. As $k \to \infty$, the limiting form of the chi-square distribution is the normal distribution.

χ^2 Distribution

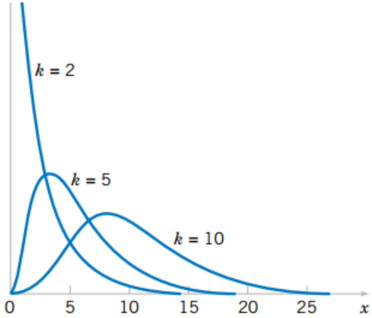

Figure 6: Probability density functions of several χ^2 distributions

χ^2 Distribution

Note

It is customary to let χ^2_α represent the χ^2 value above which we find an area of α. This is illustrated by the shaded region in Figure 7.

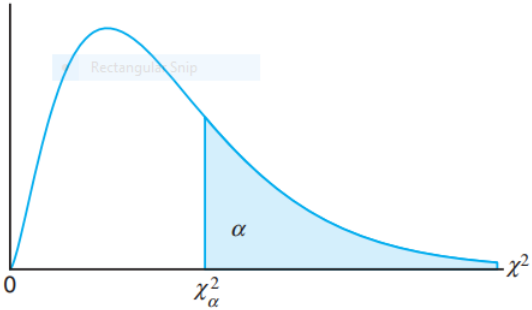

Figure 7: The chi-squared distribution

t-Distribution

Theorem 4

Let Z be a standard normal random variable and V a chi-squared random variable with ν degrees of freedom. If Z and V are independent, then the distribution of the random variable T, where

$$T = \frac{Z}{\sqrt{V/(n-1)}}$$

is given by the density function

$$h(t) = \frac{\Gamma[(\nu+1)/2]}{\Gamma(\nu/2)\sqrt{\pi\nu}}\left(1+\frac{t^2}{\nu}\right)^{-(\nu+1)/2}, \quad -\infty < t < +\infty,$$

where ν is the number of degrees of freedom. The mean and variance of the t distribution are zero and $\nu/(\nu-2)$ (for $\nu > 2$), respectively.

t-Distribution

Corollary 1

Let $X_1, X_2, \ldots, X_n$ be independent random variables that are all normal with mean μ and standard deviation σ. Let

$$\overline{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n) \quad \text{and} \quad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Then the random variable

$$\boxed{T = \frac{\overline{X} - \mu}{S/\sqrt{n}}} \tag{9}$$

has a t-distribution with $\nu = n - 1$ degrees of freedom.

t-Distribution

Note

The probability distribution of T was first published in 1908 in a paper written by W.S. Gosset. At the time, Gosset was employed by an Irish brewery that prohibited publication of research by members of its staff. To circumvent this restriction, he published his work secretly under the name "Student." Consequently, the distribution of T is usually called the Student t-distribution or simply the t-distribution.

t-Distribution

What does the t-distribution look like?

1. The distribution of T is similar to the distribution of Z in that they both are symmetric about a mean of zero.
2. Both distributions are bell shaped, but the t-distribution is more variable, owing to the fact that the t-values depend on the fluctuations of two quantities, $\overline{X}$ and S^2, whereas the Z-values depend only on the changes in $\overline{X}$ from sample to sample.
3. The distribution of T differs from that of Z in that the variance of T depends on the sample size n and is always greater than 1.
4. Only when the sample size $n \to \infty$ will the two distributions become the same.

t-Distribution

What does the t-distribution look like?

In Figure 8, we show the relationship between a standard normal distribution ($\nu = \infty$) and t-distributions with 2 and 5 degrees of freedom. The percentage points of the t-distribution are given in Table 2.

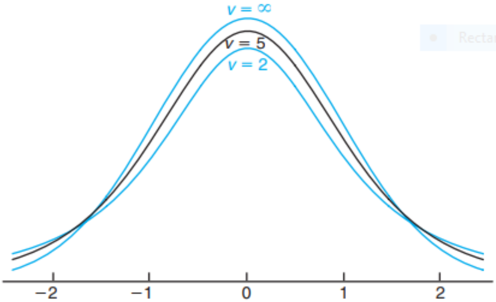

Figure 8: The t-distribution curves for $\nu = 2, 5,$ and ∞

t-Distribution

Critical Values of the t-Distribution

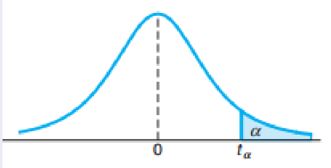

Figure 9: Critical Values of the t-Distribution

t-Distribution

Table 2

d.f. \ α	0.10	0.05	0.025	0.01	0.005	0.002	0.0005
1	3.078	6.314	12.706	31.821	63.526	318.309	363.6
2	1.886	2.920	4.303	6.965	9.925	22.327	31.600
3	1.638	2.353	3.128	4.541	5.841	10.215	12.922
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.705	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.606	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850

t-Distribution

Table 2 (continuous)

d.f. \ α	0.10	0.05	0.025	0.01	0.005	0.002	0.0005
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.796	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
$+\infty$	1.282	1.645	1.960	2.326	2.576	3.090	3.291

t-Distribution

Note

1. It is customary to let t_α represent the t-value above which we find an area equal to α.
2. The t-value with 10 degrees of freedom leaving an area of 0.025 to the right is $t = 2.228$.
3. Since the t-distribution is symmetric about a mean of zero, we have $t_{1-\alpha} = -t_\alpha$; that is, the t-value leaving an area of $1-\alpha$ to the right and therefore an area of α to the left is equal to the negative t-value that leaves an area of α in the right tail of the distribution (see Figure 10. That is, $t_{0.95} = -t_{0.05}$, $t_{0.99} = -t_{0.01}$, and so forth).

t-Distribution

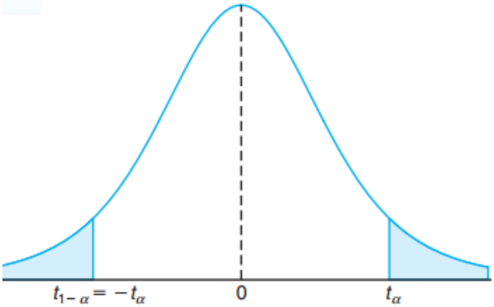

Figure 10: Symmetry property (about 0) of the t-distribution

t-Distribution

Example 14

The t-value with $\nu = 14$ degrees of freedom that leaves an area of 0.025 to the left, and therefore an area of 0.975 to the right, is

$$t_{0.975} = -t_{0.025} = -2.145$$

Example 15

Find $P([-t_{0.025} < T < t_{0.05})$.

Solution

Since $t_{0.05}$ leaves an area of 0.05 to the right, and $-t_{0.025}$ leaves an area of 0.025 to the left, we find a total area of $1 - 0.05 - 0.025 = 0.925$ between $-t_{0.025}$ and $t_{0.05}$. Hence

$$P(-t_{0.025} < T < t_{0.05}) = 0.925.$$

t-Distribution

Example 16

Find k such that $P(k < T < -1.761) = 0.045$ for a random sample of size 15 selected from a normal distribution and $\dfrac{\overline{X} - \mu}{\sigma/\sqrt{n}}$.

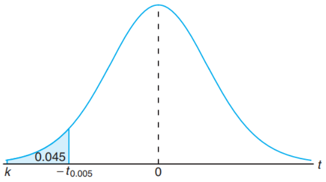

Figure 11: The t-values for Example 16

t-Distribution

Solution

From Table 2 we note that 1.761 corresponds to $t_{0.05}$ when $\nu = 14$. Therefore, $-t_{0.05} = -1.761$. Since k in the original probability statement is to the left of $-t_{0.05} = -1.761$, let $k = -t_\alpha$. Then, from Figure 11, we have $0.045 = 0.05 - \alpha$, or $\alpha = 0.005$. Hence, from Table 2 with $\nu = 14$, $k = -t_{0.005} = -2.977$ and

$$P(-2.977 < T < -1.761) = 0.045.$$

t-Distribution

Example 17

A chemical engineer claims that the population mean yield of a certain batch process is 500 grams per milliliter of raw material. To check this claim he samples 25 batches each month. If the computed t-value falls between $-t_{0.05}$ and $t_{0.05}$, he is satisfied with this claim. What conclusion should he draw from a sample that has a mean $\overline{x} = 518$ grams per milliliter and a sample standard deviation $s = 40$ grams? Assume the distribution of yields to be approximately normal.

t-Distribution

Solution

From Table 2 we find that $t_{0.05} = 1.711$ for 24 degrees of freedom. Therefore, the engineer can be satisfied with his claim if a sample of 25 batches yields a t-value between -1.711 and 1.711. If $\mu = 500$, then

$$t = \frac{518 - 500}{40/\sqrt{25}} = 2.25,$$

a value well above 1.711. The probability of obtaining a t-value, with $\nu = 24$, equal to or greater than 2.25 is approximately 0.02. If $\mu > 500$, the value of t computed from the sample is more reasonable. Hence, the engineer is likely to conclude that the process produces a better product than he thought.

t-Distribution

What is the t-distribution used for?

1. The reader should note that use of the t-distribution for the statistic

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

requires that $X_1, X_2, \ldots, X_n$ be normal.

2. The use of the t-distribution and the sample size consideration do not relate to the Central Limit Theorem. The use of the standard normal distribution rather than T for $n \geq 30$ merely implies that S is a sufficiently good estimator of σ in this case.

Content

Unbiased Estimators

Note

An estimator should be "close" in some sense to the true value of the unknown parameter. Formally, we say that is $\widehat{\Theta}$ an unbiased estimator of θ if the expected value of $\widehat{\Theta}$ is equal to θ. This is equivalent to saying that the mean of the probability distribution of $\widehat{\Theta}$ (or the mean of the sampling distribution of $\widehat{\Theta}$) is equal to θ.

Definition 9 (Bias of an Estimator)

A statistic $\hat{\Theta}$ is said to be an unbiased estimator of the parameter θ if

$$\mu_{\hat{\Theta}} = E(\hat{\Theta}) = \theta.$$

If the estimator is not unbiased, then the difference

$$\boxed{E(\widehat{\Theta}) - \theta} \tag{10}$$

is called the bias of the estimator $\widehat{\Theta}$.

Unbiased Estimators

Example 18

Show that S^2 is an unbiased estimator of the parameter σ^2.

Solution

$$E(S^2) = E\Big[\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2\Big]$$
$$= \frac{1}{n-1}\Big[\sum_{i=1}^{n}E(X_i-\mu)^2 - nE(\overline{X}-\mu)^2\Big] = \frac{1}{n-1}\Big(\sum_{i=1}^{n}\sigma^2_{X_i} - \sigma^2_{\overline{X}}\Big).$$

However,

$$\sigma^2_{X_i} = \sigma^2, \quad \text{for } i = 1, 2, \ldots, n, \quad \text{and } \sigma^2_{\overline{X}} = \frac{\sigma^2}{n}.$$

Therefore,

$$E(S^2) = \frac{1}{n-1}\Big(n\sigma^2 - n\frac{\sigma^2}{n}\Big) = \sigma^2.$$

Unbiased Estimators

Example 19

Sometimes there are several unbiased estimators of the sample population parameter. For example, suppose we take a random sample of size $n = 10$ from a normal population and obtain the data $x_1 = 12.8$, $x_2 = 9.4$, $x_3 = 8.7$, $x_4 = 11.6$, $x_5 = 13.1$, $x_6 = 9.8$, $x_7 = 14.1$, $x_8 = 8.5$, $x_9 = 12.1$, $x_{10} = 10.3$. Now the sample mean is

$$\overline{x} = \frac{12.8 + 9.4 + 8.7 + 11.6 + 13.1 + 9.8 + 14.1 + 8.5 + 12.1 + 10.3}{10} = 11.04,$$

the sample median is

$$\widehat{x} = \frac{10.3 + 11.6}{2} = 10.95$$

and a 10% trimmed mean (obtained by discarding the smallest and largest 10% of the sample before averaging) is

$$\overline{x}_{tr(10)} = \frac{8.7 + 9.4 + 9.8 + 10.3 + 11.6 + 12.1 + 12.8 + 13.1}{8} = 10.98.$$

We can show that all of these are unbiased estimates of μ.

Variance of a Point Estimator

Since there is not a unique unbiased estimator, we cannot rely on the property of unbiasedness alone to select our estimator. We need a method to select among unbiased estimators.
Suppose that $\widehat{\Theta}_1$ and $\widehat{\Theta}_2$ are unbiased estimators of θ, $E(\widehat{\Theta}_1) = \theta$ and $E(\widehat{\Theta}_2) = \theta$. However, the variance of these distributions may be different (see Fig. 12).

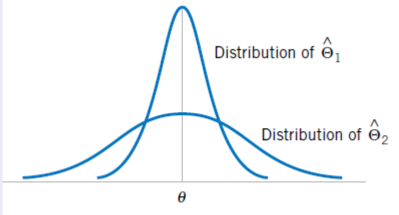

Figure 12: The sampling distributions of two unbiased estimators $\widehat{\Theta}_1$ and $\widehat{\Theta}_2$

Since $\widehat{\Theta}_1$ has a smaller variance than $\widehat{\Theta}_2$, the estimator $\widehat{\Theta}_1$ is more likely to produce an estimate close to the true value θ.

Variance of a Point Estimator

Definition 10 (Minimum Variance Unbiased Estimator)

If we consider all unbiased estimators of θ, the one with the smallest variance is called the minimum variance unbiased estimator (MVUE).

Theorem 5

If $(X_1, X_2, \ldots, X_n)$ is a random sample of size n from a normal distribution with mean μ and variance σ^2, the sample mean $\overline{X}$ is the MVUE for μ.

Variance of a Point Estimator

Definition 11 (Most efficient estimator)

If we consider all possible unbiased estimators of some parameter θ, the one with the smallest variance is called the most efficient estimator of θ.

Figure

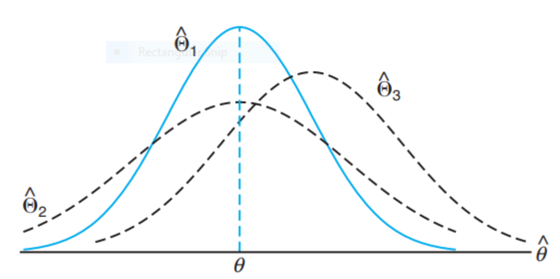

Figure 13: Sampling distributions of different estimators of θ

Variance of a Point Estimator

Note

Figure 13 illustrates the sampling distributions of three different estimators, $\hat{\Theta}_1$, $\hat{\Theta}_2$, and $\hat{\Theta}_3$, all estimating θ. It is clear that only $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are unbiased, since their distributions are centered at θ. The estimator $\hat{\Theta}_1$ has a smaller variance than $\hat{\Theta}_2$ and is therefore more efficient. Hence, our choice for an estimator of θ, among the three considered, would be $\hat{\Theta}_1$.

Standard Error

Definition 12 (Standard Error of an Estimator)

The standard error of an estimator $\widehat{\Theta}$ is its standard deviation, given by $\sigma_{\widehat{\Theta}} = \sqrt{V(\widehat{\Theta})}$. If the standard error involves unknown parameters that can be estimated, substitution of those values into $\sigma_{\widehat{\Theta}}$ produces an estimated standard error, denoted by $\widehat{\sigma}_{\widehat{\Theta}}$.

Standard Error

Example 20

Suppose we are sampling from a normal distribution with mean μ and variance σ^2. Now the distribution of $\overline{X}$ is normal with mean μ and variance σ^2/n, so the standard error of $\overline{X}$ is

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}.$$

If we did not know σ but substituted the sample standard deviation S into the above equation, the estimated standard error of $\overline{X}$ would be

$$\widehat{\sigma}_{\overline{X}} = \frac{S}{\sqrt{n}}.$$

Mean Squared Error of an Estimator

Sometimes it is necessary to use a biased estimator. In such cases, the mean squared error of the estimator can be important. The mean squared error of an estimator $\widehat{\Theta}$ is the expected squared difference between $\widehat{\Theta}$ and θ.

Definition 13 (Mean Squared Error of an Estimator)

The mean squared error of an estimator $\widehat{\Theta}$ of the parameter θ is defined as

$$\boxed{\mathrm{MSE}(\widehat{\Theta}) = E(\widehat{\Theta} - \theta)^2} \tag{11}$$